Task 1

Create a Python program that:
Prints a welcome message.
Take your Name, College Name, and Branch as input, Displays the entered information in a formatted output.

Source Code:

```
print("Welcome to Python Programming ")
name=str(input("Enter Your Name : "))
college=str(input("Enter your College Name : "))
branch =str(input("Enter your Branch : "))
print("\n\n")

print("Welcome",name)
print("From College : ",college)
print("Your Branch is : ",branch)
```

Output :

```
PS C:\Users\rahul\OneDrive\Documents\Python Fundamentals> python -u "c:\Users\r
Welcome to Python Programming
Enter Your Name : Shyam Singh
Enter your College Name : RATM
Enter your Branch : Computer Science



Welcome Shyam Singh
From College :  RATM
Your Branch is :  Computer Science
PS C:\Users\rahul\OneDrive\Documents\Python Fundamentals>
```

Explanation

First, I printed "Welcome to Python Programming" on the screen. After that, I took input from the user, including their name, college name, and branch. Finally, I printed all three details entered by the user.

Task 2

Write a Python program that:
Declares variables of different data types:
Integer
Float
String
Boolean
Print each variable along with its data type using type().

Source Code:

```
a = int(input("Enter a Integer value: "))
b = float(input("Enter a float value: "))
c = input("Enter a string: ")
d = True
print("\n\n")

print("Value:", a, "and Data Type:", type(a))
print("Value:", b, "and Data Type:", type(b))
print("Value:", c, "and Data Type:", type(c))
print("Value:", d, "and Data Type:", type(d))
```

Output:

```
PS C:\Users\rahul\OneDrive\Documents\Python Fundamentals> python -u "
Enter a Integer value: 34
Enter a float value: 23.4
Enter a string: rahul



Value: 34 and Data Type: <class 'int'>
Value: 23.4 and Data Type: <class 'float'>
Value: rahul and Data Type: <class 'str'>
Value: True and Data Type: <class 'bool'>
PS C:\Users\rahul\OneDrive\Documents\Python Fundamentals>
```

Explanation

First, I took an integer value, a float value, and a string as input from the user. I also assigned a Boolean value (True) to a variable. After that, I printed all four values along with their respective data types using the type() function.

Task 3

Create a calculator program that performs:
Addition
Subtraction
Multiplication
Division
Modulus
using two numbers entered by the user.

Source Code:

```
num1=int(input("Enter First Number : "))
num2=int(input("Enter Second Number : "))
print("\n\n")

print("Addition of",num1,"and",num2,"is : ",num1+num2)
print("Subtraction of",num1,"and",num2,"is : ",num1-num2)
print("Multiplication of",num1,"and",num2,"is : ",num1*num2)
print("Division of",num1,"and",num2,"is : ",num1/num2)
print("Modulus of",num1,"and",num2,"is : ",num1%num2)
```

Output ;

```
Enter First Number : 34
Enter Second Number : 17



Addition of 34 and 17 is :  51
Subtraction of 34 and 17 is :  17
Multiplication of 34 and 17 is :  578
Division of 34 and 17 is :  2.0
Modulus of 34 and 17 is :  0
```

Explanation

First, I took two integer numbers as input from the user. Then, I performed five arithmetic operations on these numbers: addition, subtraction, multiplication, division, and modulus. Finally, I displayed the results of all five operations using the print() function.

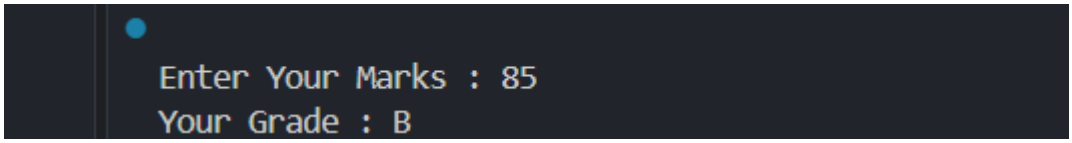
Task 4

Write a Python program that:
Takes marks as input.
Displays the grade using if, elif, and else.
Example:
90+ → A
75–89 → B
60–74 → C
Below 60 → Fail

Source Code:

```
marks = int(input("Enter Your Marks : "))
if marks >= 90:
    print("Congratulations! Your Grade : A")
elif marks >= 75:
    print("Your Grade : B")
elif marks >= 60:
    print("Your Grade : C")
else:
    print("Your Grade : Fail")
```

Output:



Explanation:

First, I took the student's marks as input from the user. Then, I used if, elif, else statements to check the marks and assign a grade accordingly. If the marks were 90 or above, the grade was A; if they were 75 or above, the grade was B; if they were 60 or above, the grade was C. Otherwise, the program displayed Fail. Finally, I printed the grade based on the marks entered by the user.

Task 5

Write Python programs to:

Print numbers from 1 to 20 using a for loop.

Print the multiplication table of any number.

Print even numbers from 1 to 50 using a while loop.

Source Code:

```
num1=int(input("Enter The Number : "))
print("Print numbers from 1 to", num1)
for i in range(1, num1+1):
    print(i, end="\n")
print("\n")
print("Table of", num1)
for i in range(1, 11):
    print( num1*i, end="\n")
print("\n")
print("Even Numbers from 1 to", num1)
for i in range(1, num1+1):
    if(i%2==0):
        print(i, end="\n")
```

Output:

```
PS C:\Users\rahul\OneDrive\Documents\Python Fundamentals>
Enter The Number : 20
Print numbers from 1 to 20
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20


Table of 20
20
40
60
80
100
120
140
160
180
200


Even Numbers from 1 to 20
2
4
6
8
10
12
14
16
18
20
```

Explanation :

First, I took a number as input from the user. Then, I printed numbers from 1 to that number, displayed its multiplication table, and printed all even numbers in the same range using `for` loops and an `if` condition.

Task 6

Create two user-defined functions:
Function to calculate the square of a number.
Function to calculate the average of three numbers.
Call both functions with user input.

Source Code:

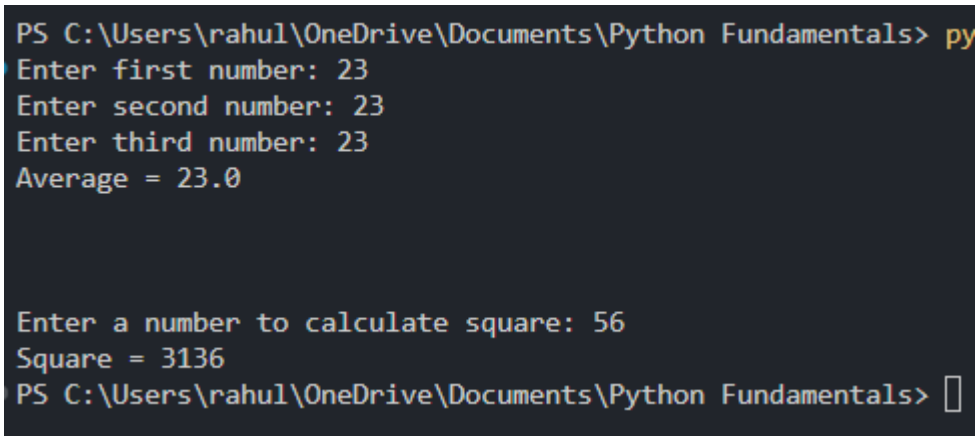
```
def average(num1, num2, num3):
    return (num1 + num2 + num3) / 3

def square(num):
    return num * num

num1 = float(input("Enter first number: "))
num2 = float(input("Enter second number: "))
num3 = float(input("Enter third number: "))
print("Average =", average(num1, num2, num3))

print("\n\n")
num = int(input("Enter a number to calculate square: "))
print("Square =", square(num))
```

Output:



Explanation :
First, I created two functions (i) average() to calculate the average of three numbers and (ii) square() to calculate the square of a number. Then, I took three numbers as input from the user and passed them into the average() function to calculate and display their average. After that, I took an integer as input, passed it to the square() function, and displayed its square. This task helped me understand how to define and call functions, pass arguments, and return values in Python.

Task 7

Create programs to demonstrate:
String operations (upper(), lower(), replace(), find())
List operations (append(), remove(), sort())
Tuple creation and indexing
Dictionary storing student information
Set operations (add(), remove())

Source Code:

```
text = "Hello Rahul Saini"
print("Original Text:", text)
print("Uppercase:", text.upper())
print("Lowercase:", text.lower())
print("Replace:", text.replace("Saini", "Singh"))
print("Find:", text.find("Rahul"))
print("\n\n")

numbers = [40, 10, 30, 20]
print("List of Numbers:", numbers)
numbers.append(50)
print("After Append:", numbers)
numbers.remove(10)
print("After Remove:", numbers)
numbers.sort()
print("After Sort:", numbers)
print("\n\n")

fruits = ("Apple", "Banana", "Mango")
print("Tuple:", fruits)
print("First Element:", fruits[0])
print("Second Element:", fruits[1])
print("Third Element:", fruits[2])
print("\n\n")
```

Output:

```
Original Text: Hello Rahul Saini
Uppercase: HELLO RAHUL SAINI
Lowercase: hello rahul saini
Replace: Hello Rahul Singh
Find: 6



List of Numbers: [40, 10, 30, 20]
After Append: [40, 10, 30, 20, 50]
After Remove: [40, 30, 20, 50]
After Sort: [20, 30, 40, 50]



Tuple: ('Apple', 'Banana', 'Mango')
First Element: Apple
Second Element: Banana
Third Element: Mango



Student Information: {'Name': 'Shyam Singh', 'Branch': 'Computer Science', 'College': 'ABC College'}
Student Name: Shyam Singh
Branch: Computer Science



Set of Colors: {'Green', 'Red', 'Blue'}
After Add: {'Green', 'Red', 'Blue', 'Yellow'}
After Remove: {'Red', 'Blue', 'Yellow'}
PS C:\Users\rahul\OneDrive\Documents\Python Fundamentals>
```

Explanation

First, I performed different string operations, such as converting text to uppercase and lowercase, replacing a word, and finding a word's position. Then, I created a list of numbers and used append() to add a number in list, remove() to delete a number from list, and sort() to modify and arrange the list. Finally, I created a tuple of fruits and accessed its elements using indexing. This task helped me understand basic string operations and the use of lists and tuples in Python.

Task 8

Write a Python program that:

Creates a text file.
Write your introduction into the file.
Reads and displays the file contents.

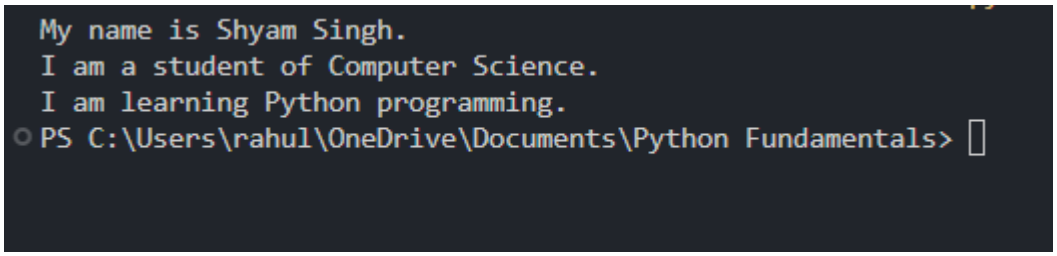
Source Code:

```
#To create a text file and write Introduction
file = open("introduction.txt", "w")
file.write("My name is Shyam Singh.\n")
file.write("I am a student of Computer Science.\n")
file.write("I am learning Python programming.")

file.close()
# To read and display file
file = open("introduction.txt", "r")

content = file.read()
print(content)
file.close()
```

Output:



Explanation :

First, I created a text file named introduction.txt using the open() function in write mode (“w”).
Then, I wrote my introduction into the file using the write() method.
I used \n to write each sentence on a new line.
After writing the content, I closed the file using close()
Next, I opened the same file in read mode (“r”).
I used the read() method to read the complete content of the file.
Finally, I displayed the content using the print() function and closed the file again.
This task helped me understand how to create, write, and read text files in Python.

Task 9

Develop a Student Record Management System using Python.
The program should:
Add student details.
Display all student records.
Search for a student by name.
Delete a student record.
Use lists or dictionaries for data storage.

Source Code:

```
students = []
while True:
    print("\n1. Add Student")
    print("2. Display Students")
    print("3. Search Student")
    print("4. Delete Student")
    print("5. Exit")

    choice = input("Enter your choice: ")

    if choice == "1":
        name = input("Enter name: ")
        roll = input("Enter roll number: ")
        branch = input("Enter branch: ")

        student = {
            "name": name,
            "roll": roll,
            "branch": branch
        }

        students.append(student)
        print("Student added successfully")

    elif choice == "2":
        if len(students) == 0:
            print("No records available")
        else:
            for s in students:
                print("Name:", s["name"])
                print("Roll Number:", s["roll"])
                print("Branch:", s["branch"])
                print("----------------")

    elif choice == "3":
```

```
        name = input("Enter name to search: ")
        found = False

        for s in students:
            if s["name"].lower() == name.lower():
                print("Student found")
                print("Name:", s["name"])
                print("Roll Number:", s["roll"])
                print("Branch:", s["branch"])
                found = True

        if found == False:
            print("Student not found")

    elif choice == "4":
        roll = input("Enter roll number to delete: ")
        found = False

        for s in students:
            if s["roll"] == roll:
                students.remove(s)
                print("Student deleted successfully")
                found = True
                break

        if found == False:
            print("Student not found")

    elif choice == "5":
        print("Program ended")
        break

    else:
        print("Wrong choice")
```

Output:

```
PS C:\Users\rahul\OneDrive\Documents\Python Fundamentals> python -u

1. Add Student
2. Display Students
3. Search Student
4. Delete Student
5. Exit
Enter your choice: 1
Enter name: Shyam Singh
Enter roll number: 25
Enter branch: CS
Student added successfully

1. Add Student
2. Display Students
3. Search Student
4. Delete Student
5. Exit
Enter your choice: 2
Name: Shyam Singh
Roll Number: 25
Branch: CS
---------------

1. Add Student
2. Display Students
3. Search Student
4. Delete Student
5. Exit
Enter your choice: 5
Program ended
PS C:\Users\rahul\OneDrive\Documents\Python Fundamentals>
```

Explanation :

First, I created an empty list to store student details.
I used a while loop to keep the program running.
I displayed a menu with options to add, display, search, delete, and exit.
I took the user's choice as input.
For adding a student, I took their name, roll number, and branch.
I stored these details in a dictionary and added it to the list.
For displaying students, I checked whether records were available.
I used a for loop to display all student details.
For searching, I asked the user to enter the student's name.
I used a loop to find the student in the list.
I used the lower() method to make the search case-insensitive.
If the student was found, I displayed their details.
For deleting a student, I asked for their roll number.
I searched for the matching roll number and removed the record.
I used if-elif-else statements to handle different choices.
I used a break to exit the program.
This task helped me learn how to manage student records using Python.